



Chapter 1:

Average velocity = $\bar{v} = \frac{1}{T} \int_0^T v(t) dt = \frac{1}{T} \Delta x$
"always time averaged"

Instantaneous velocity, $\vec{v} = \frac{d\vec{x}}{dt}$

Acceleration = $\frac{d\vec{v}}{dt}$

If $\vec{a}$ is linear, j is constant

$\vec{a} = \frac{d\vec{v}}{dt}$

$\vec{a} = \frac{d|\vec{v}| \cdot \hat{v}}{dt}$
 $= |\vec{v}| \frac{d\hat{v}}{dt} + \hat{v} \frac{d|\vec{v}|}{dt}$

When $\vec{a}$ is constant,

$\frac{d\vec{a}}{dt} = \vec{j}$

$\int \vec{a} dt = \int \frac{d\vec{v}}{dt} dt$

$\int \frac{d\vec{v}}{dt} dt = \vec{j} dt$

$\vec{a}t + \vec{v}_0 = \vec{v}$ or
 $\Delta \vec{v} = \Delta \vec{v}$

$\vec{a} = \vec{j}t + \vec{a}_0$ or
 $\Delta \vec{a} = \vec{j}t$

$\int \vec{a}t + \vec{v}_0 dt = \int \vec{j} dt$

$\int \vec{a} dt = \int \vec{j}t + \vec{a}_0 dt$

$\frac{\vec{a}t^2}{2} + \vec{v}_0t + \vec{v}_0 = \vec{v}$ or
 $\Delta \vec{v} = \vec{v}_0t + \frac{1}{2}\vec{a}t^2$

$\vec{v} = \frac{\vec{j}t^2}{2} + \vec{a}_0t + \vec{v}_0$ or
 $\Delta \vec{v} = \vec{j}\frac{t^2}{2} + \vec{a}_0t$

$\int \vec{v} dt = \int \frac{\vec{j}t^2}{2} + \vec{a}_0t + \vec{v}_0 dt$

$\vec{r} = \frac{\vec{j}t^3}{6} + \frac{\vec{a}_0t^2}{2} + \vec{v}_0t + \vec{r}_0$ or
 $\Delta \vec{r} = \vec{j}\frac{t^3}{6} + \frac{\vec{a}_0t^2}{2} + \vec{v}_0t$

Chapter 2:

Momentum: $\vec{p} = m \times \vec{v}$

$\frac{d\vec{p}}{dt} = m \times \frac{d\vec{v}}{dt}$

$\frac{d\vec{p}}{dt} = m \times \vec{a}$

$= \vec{F}_{net}$, if m is constant

if we break calculus,

$\frac{\Delta \vec{p}}{\Delta t} = \vec{F}_{net}$

$\Delta \vec{p} = \vec{F}_{net} \Delta t$ impulse, if Δt is small s.t.
 $\vec{F}_{net} \sim \text{constant}$

$\vec{J} = \int \vec{F} dt$

$\vec{r}_f = \vec{r}_i + \vec{v} \cdot \Delta t$
 $\vec{r}_f = \vec{r}_i + \frac{\vec{p}}{m} \Delta t$

In modelling questions,

*Never directly assume the value of time,
 instead determine via average velocity and Δx just "assumed" if $\vec{F}_{net}$ is constant*

When predicting motion:

① $\vec{F}_{net} = \sum \vec{F}_i$ *← Find $\vec{F}_{net}$*

② $\vec{p}_f = \vec{p}_i + \vec{F}_{net} \Delta t$ *← Update $\vec{p}$*

③ $\vec{r}_f = \vec{r}_i + \frac{\vec{p}}{m} \Delta t$ *← Find position*

Uncertainties,

- Type A statistically measured, uncontrollable variations
- Type B based on a device, estimated to model the outputs of a device

Type A:

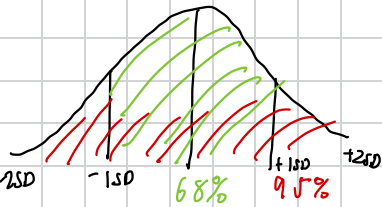
$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i \quad \sigma^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2 \quad u(\bar{x}) = \frac{\sigma}{\sqrt{N}}$$

Type B: firm limit of device (it can never be worse than $\pm a$)

$\hookrightarrow a/2$

$\hookrightarrow$ For example, laser ruler, $a = 0.5 \text{ mm}$, $\frac{a}{2} = 0.25 \text{ mm}$

Normal distribution:



Chapter 3:

Binomial approximation (holds for small x)

$$(1+x)^n \approx (1+nx)$$

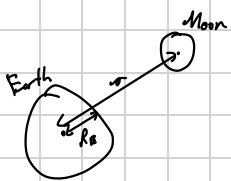
$$|\vec{F}_g| = \frac{m G M_E}{(R_E + y)^2} = \frac{m M_E G}{R_E^2 (1 + \frac{y}{R_E})^2} \approx \frac{m M_E G}{R_E^2} (1 - 2 \frac{y}{R_E}) = mg (1 - 2 \frac{y}{R_E})$$

$\hookrightarrow$ Flat Earth Approximation

since height is small, binomial approximation can be applied to the ellipse so trajectory that it is effectively $\sim$ a parabola

- Treating Earth as a point mass

Limitation: No tidal force.



$$\begin{aligned} \Delta g &= \frac{G M_E}{r^2} - \frac{G M_E}{(r - R_E)^2} \\ &= \frac{G M_E}{r^2} - \frac{G M_E}{r^2 (1 - \frac{R_E}{r})^2} \\ &= \frac{G M_E}{r^2} \left(1 - \frac{1}{(1 - \frac{R_E}{r})^2} \right) \\ g &= \frac{G M_E}{r^2} \end{aligned}$$

$$\begin{aligned} \frac{\Delta g}{g} &= 1 - \frac{1}{(1 - \frac{R_E}{r})^2} \\ &\approx 1 - (1 + 2 \frac{R_E}{r}) \\ &= - \frac{2 R_E}{r} \end{aligned}$$

- Center of mass: $\vec{r}_{cm} = \frac{\sum \vec{r}_{ci} m_{ci}}{M_{tot}}$

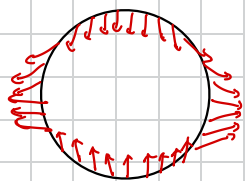
$\hookrightarrow$ More accurately:

$$\vec{r}_{cm} = \frac{1}{M} \int \vec{r} \rho(\vec{r}) dV$$

- Orbits are binomial ellipses as opposed to parabolas.

Issue with Flat Earth approximation, tidal force.

↳ Tidal force is the difference of gravitational force on the surface of the Earth and the center of the Earth.



The difference:

$$\frac{\Delta g}{g} \approx -\frac{2R_E}{r}$$

Circular orbits:

$$v^2 \propto \frac{M(r)}{r} \leftarrow \text{Mass as a f of radius}$$

Chapter 4

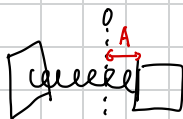
Simple harmonic motion:

Restoring force: $\frac{dp_x}{dt} \approx m \frac{d^2x}{dt^2} = -k(x-L)$, if rest $\neq$ origin.

Position: $x(t) = A \cos(\omega t + \phi_0) + L$

Velocity: $v(t) = -A\omega \sin(\omega t + \phi_0)$

Acceleration: $a(t) = -A\omega^2 \cos(\omega t + \phi_0)$
 $= -\omega^2 x$



Important variables:

k = spring constant / stiffness

ω = angular frequency

$$= \sqrt{\frac{k}{m}}$$

T = period

$$= \frac{2\pi}{\omega}$$

f = frequency

$$= \frac{1}{T} \text{ or } \frac{\omega}{2\pi}$$

A = amplitude

ϕ = phase shift

Some useful equations:

Maximum speed: $|v| = \pm A\omega$, $x=0$

Maximum acceleration: $a = \pm A\omega^2$, $x = \pm A$

Average speed:

$$|\bar{v}| = \frac{2A\omega}{\pi} = \frac{2}{\pi} |v|, |v| \text{ is maximum speed}$$

Average distance from equilibrium:

$$|\bar{x}| = \frac{2A}{\pi}$$

Equate acceleration and restoring force:

$$-kx = -m\omega^2 x$$

$$k = m\omega^2$$

$$\omega^2 = \frac{k}{m}$$

Pendulum and harmonic oscillator:

$$\theta(t) = \overset{\text{or } A}{\theta_0} \cos\left[2\pi \frac{t}{T} + \phi_0\right]$$

θ_0 = amplitude

$$\omega^2 = \frac{g}{L}$$

$$T \approx 2\pi \sqrt{\frac{L}{g}}$$

Ball spring model of solids: (only ρ and E are macroscopic)

1. Determining d , separation distance of atomic bonds:

$\hookrightarrow$ nucleon $\sim 1.67 \times 10^{-27} \text{ kg}$

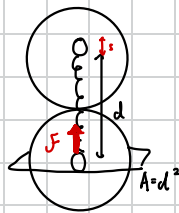
$\hookrightarrow$ mass, $m = (\# \text{ protons, neutrons}) (m_{\text{nucleon}})$



$\hookrightarrow$ density, $\rho = \frac{m}{v}$, assume the atom is a cube

$$\rho = \frac{m}{d^3}$$

$$d = \sqrt[3]{\frac{m}{\rho}}$$



s = microscopic displacement
 d = distance
 F = microscopic spring force.

2. Determining k , spring constant of solid:

$\hookrightarrow F = ks$, k is the microscopic spring constant

$\hookrightarrow E$ (macroscopic) = $\frac{\text{stress } (\sigma)}{\text{strain } (\epsilon)}$

$$= \frac{ks/d^2}{s/d} = \frac{k}{d}$$

$\therefore E = \frac{k}{d}$, $k = Ed$

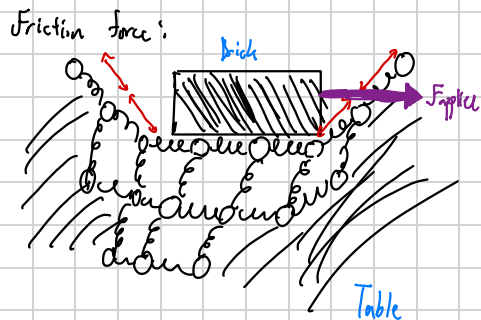
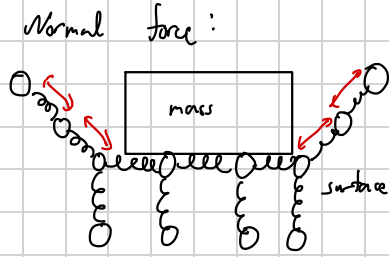
3. We know $v \sim \text{frequency} \cdot d$

With respect to angular frequency,

$$v \sim \omega d$$

$$\sim \sqrt{\frac{E}{\rho}}$$

4. Speed of sound in solid $\sim \sqrt{\frac{E}{\rho}}$, macroscopic quantities



$\hookrightarrow$ Kinetic friction: "almost all material has a higher $\mu_s \geq \mu_k$, for some"

$$f_k = \mu_k N$$

$\hookrightarrow$ Static friction: $\mu_s = \mu_k$

$$f_s \leq \mu_s N$$

$\hookrightarrow$ Static friction can be any direction

Chapter 5:

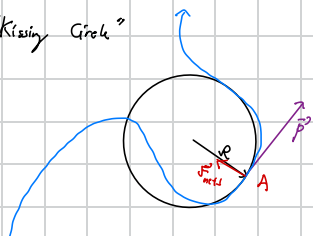
Centripetal force:

↳ The $\perp$ component of net force

$$F_{\text{net } \perp} = \frac{mv^2}{R}, \quad R \text{ is the radius of "kissing circle"}$$

↳ associated with the directional change ($\Delta \hat{p}$)

"Kissing Circle"



$$\vec{p} = \hat{p} |\vec{p}|$$

direction magnitude

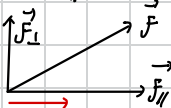
$$\frac{d\vec{p}}{dt} = \frac{d\hat{p}}{dt} |\vec{p}| + \frac{d|\vec{p}|}{dt} \hat{p}$$

$$\Delta F_{\text{net}} = \Delta F_{\text{net } \perp} + \Delta F_{\text{net } \parallel}$$

↳ Note: $F_{\parallel}$ is the parallel component of net force

↳ associated with magnitude change ($|\vec{p}|$)

$$\vec{F} = \vec{F}_{\parallel} + \vec{F}_{\perp}$$

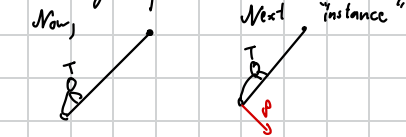


Rate of change of direction:

$$\left| \frac{d\hat{p}}{dt} \right| = \frac{v}{R} = \omega$$

Centripetal force & "pendulums":

Turning points:

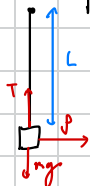


when T is momentarily at rest,

$$\vec{p} = 0$$

$$\frac{d\vec{p}}{dt}, \vec{F}_{\text{net}} \neq 0$$

Bottom point:



$F_{\text{net } \parallel} = 0$ (no horizontal force)

$$F_{\text{net}} = F_{\text{net } \perp}$$

$$= T - mg$$

$$T - mg = \frac{mv^2}{R}$$

$$T = \frac{mv^2}{R} + mg$$

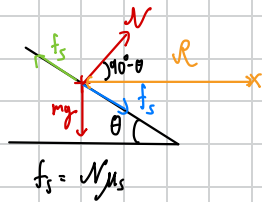
Tension is much greater as:

if $p = 0$,

$$T = mg$$

Centripetal force and banked curves

When speed is high,
When speed is low,



$$f_s = N \mu_s$$

For max speed,

$$\sum F_y = 0$$

$$N \cos \theta = mg + f_s \sin \theta$$

$$N \cos \theta = mg + N \mu_s \sin \theta$$

$$mg = N (\cos \theta - \mu_s \sin \theta) \quad (1)$$

$$\sum F_x = \frac{mv^2}{R}$$

$$N \sin \theta + f_s \cos \theta = \frac{mv^2}{R}$$

$$N \sin \theta + N \mu_s \cos \theta = \frac{mv^2}{R}$$

$$\frac{mv^2}{R} = N (\sin \theta + \mu_s \cos \theta) \quad (2)$$

For min speed,

$$\sum F_y = 0$$

$$N \cos \theta + f_s \sin \theta = mg$$

$$mg = N (\cos \theta + \mu_s \sin \theta) \quad (1)$$

$$\sum F_x = \frac{mv^2}{R}$$

$$N \sin \theta - f_s \cos \theta = \frac{mv^2}{R}$$

$$\frac{mv^2}{R} = N (\sin \theta - \mu_s \cos \theta) \quad (2)$$

Ideal case, $f_s = 0$

$$\sum F_y = 0$$

$$N \cos \theta = mg$$

$$N = \frac{mg}{\cos \theta} \quad (1)$$

$$\sum F_x = \frac{mv^2}{R}$$

$$N \sin \theta = \frac{mv^2}{R} \quad (2)$$

(2)/(1):

$$\frac{v^2}{R} = \frac{\sin \theta + \mu_s \cos \theta}{\cos \theta - \mu_s \sin \theta}$$

$$v^2 = gR \left(\frac{\sin \theta + \mu_s \cos \theta}{\cos \theta - \mu_s \sin \theta} \right)$$

(2)/(1):

$$\frac{v^2}{R} = \frac{\sin \theta - \mu_s \cos \theta}{\cos \theta + \mu_s \sin \theta}$$

$$v^2 = gR \left(\frac{\sin \theta - \mu_s \cos \theta}{\cos \theta + \mu_s \sin \theta} \right)$$

Sub (1) into (2),

$$mg \tan \theta = \frac{mv^2}{R}$$

$$v^2 = Rg \tan \theta$$

Chapter 6:

Energy conservation

↳ If a system gains energy, its surroundings must lose energy

Work:

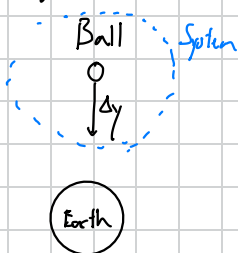
$$W = \int \vec{F} \cdot d\vec{r} = \Delta E$$

dot product

Potential energy:

$$\Delta U = -W = -\int \vec{F} \cdot d\vec{r}$$

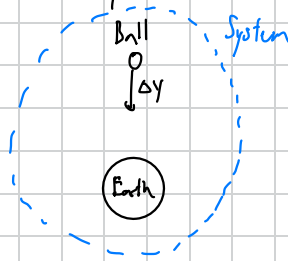
Using work:



$$\Delta K > 0$$

as $W > 0$

Using potential energy



$$\Delta K > 0$$

as $\Delta K + \Delta U = 0$
 $\Delta U = -W$
 $\Delta K = W$

↳ $\vec{F}$ must be an external force

↳ Internal interaction with the system

↳ $\vec{F}$ must be an internal force

Formula for E:

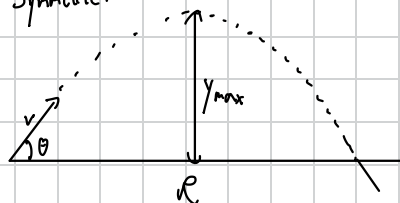
$$KE = \frac{1}{2}mv^2$$

GPE = mgh , h is arbitrary, you can define it yourself.

$$\begin{aligned} \text{Spring energy (linear)} &= \int F \, ds \\ &= \int k s \, ds \\ &= \frac{1}{2} k s^2 \end{aligned}$$

Projectile motion Review:

Symmetric:



$$y = x \tan \theta - \frac{g x^2}{2 v^2 \cos^2 \theta}, \text{ dependant of } x, \theta, v$$

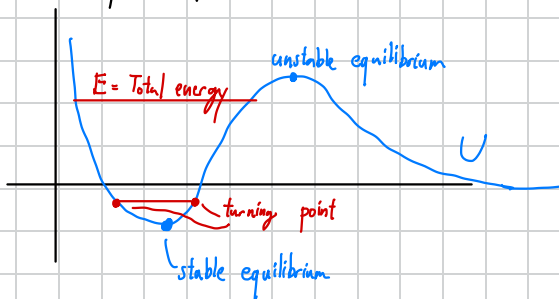
$$R = \frac{v^2 \sin 2\theta}{g}, \text{ dependant of } \theta, v$$

$$y_{\max} = \frac{v^2 \sin^2 \theta}{2g}, \text{ dependant of } \theta, v$$

$$t \text{ of flight} = \frac{2(v \sin \theta)}{g}, \text{ dependant of } v, \theta$$

$$\theta = \frac{1}{2} \arcsin\left(\frac{gR}{v^2}\right), \text{ dependant of } R, v$$

Energy Graphs:



$$\hookrightarrow K \text{ is always } \geq 0$$

$v^2 \geq 0$

$$\hookrightarrow \text{Since } \Delta U = -\int \vec{F} \cdot d\vec{r}$$

$$\frac{dU}{dr} = -\vec{F}$$

$$\hookrightarrow E = U + K$$

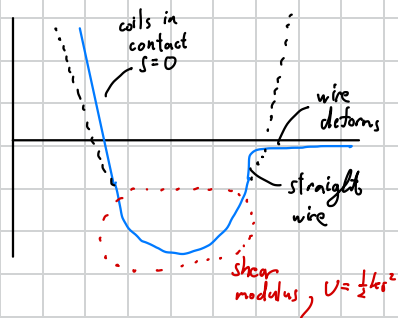
Properties:

↳ 45° gives max range

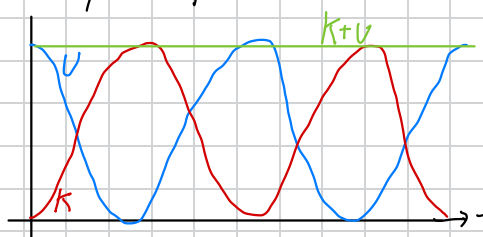
↳ 90° gives max $y_{\max}$

Chapter 7:

Potential energy graph of an actual spring:



Energy in springs graph:



Energy as heat:

↳ $E = mc\Delta T$, c is the specific heat capacity

ex. 42J is required to rise 1g of $H_2O(l)$ by $1^\circ C$

↳ Temperature = measurement of average KE in particles

Power:

↳ rate of change in E .

$$\hookrightarrow P = \frac{dE}{dt} \approx \vec{F} \cdot \vec{v}$$

$$\hookrightarrow E = \int P dt$$

↳ if only KE is present,

$$K = Pt$$

$$\frac{1}{2}mv^2 = Pt$$

$$v^2 = \frac{2Pt}{m}$$

Dissipation and Drag:

$$\vec{F}_{\text{drag}} = -\frac{1}{2}\rho_{\text{air}} C_D A v^2 \hat{v}$$

C_D , drag coefficient

A , cross-sectional area

ρ_{air} , density of air

Chapter 8:

Electron Volt (eV):

$\hookrightarrow 1\text{eV} = 1.6 \times 10^{-19}\text{J}$

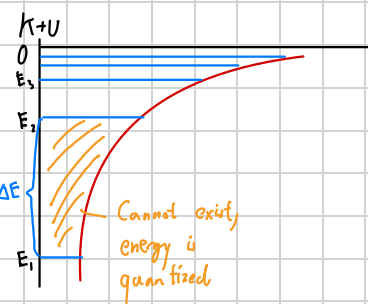
$\hookrightarrow$ Definition: energy of 1 electron accelerated through 1 volt (ie. a battery)

$\hookrightarrow$ Visible photons $\sim 2\text{eV}$

$E_{\text{red}} \sim 1\text{eV}$

$E_{\text{vio}} \sim 3\text{eV}$

Quantized energy graphs:



Example, for Hydrogen:

$E_n = \frac{-13.6}{n^2}$

$\Delta E_{1,2} = \frac{-13.6}{2^2} - \frac{-13.6}{1^2}$

$\hookrightarrow$ Discrete values $\Rightarrow$ Only allows specific radii

$\hookrightarrow$ e⁻ "jumping" or "dropping" changing radius

Boltzmann Factor:

- Boltzmann constant: $k_B = 8.6 \times 10^{-5} \text{eV/K}$

- The probability of an atom being in a state with energy E above the ground state is proportional to the Boltzmann Factor:

$\hookrightarrow P = e^{-\Delta E/k_B T}$

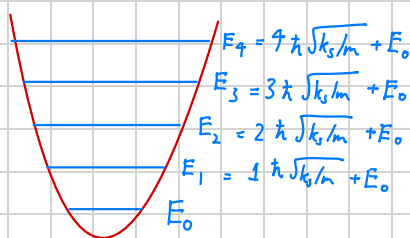
$\hookrightarrow \frac{\text{High energy springs (bonds)}}{\text{Low energy springs (bonds)}} = e^{-\Delta E/k_B T}$

Example:



$\hookrightarrow$ Each atom has 3 spring (6 half springs)

Quantized Vibrational energy:



$U(x) = \left(N + \frac{1}{2}\right) \hbar \sqrt{\frac{k_s}{m}}$

$\Delta U(x) = (N_1 - N_0) \hbar \sqrt{\frac{k_s}{m}}$

- k_s is microscopic spring constant found in Chapter 4

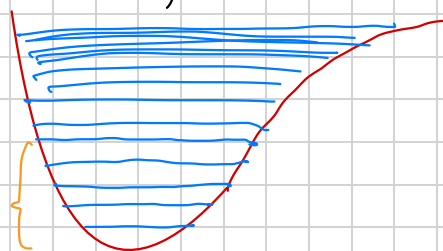
- $\hbar$, Planck's Constant $= 6.6 \times 10^{-34} \text{J.s}$

- $\hbar = \hbar / 2\pi \sim 1.05 \times 10^{-34} \text{J.s}$

- m is mass of one atom

- $E_0 = \frac{1}{2} \hbar \sqrt{\frac{k_s}{m}}$, not 0, known as "ground state"

Realistic energy model:



accurate here, near uniformly spaced.

Chapter 9

Rotational kinetic energy:

$$K_{\text{rot}} = \frac{1}{2} I \omega^2$$

↳ $\omega = \frac{2\pi}{T}$ or $\omega = \frac{v}{R}$, R = radius of object

↳ I , second moment of inertia

$$= \sum m_i r_i^2 \perp = \int \Delta M r^2$$

"summing up r_i^2 "



↳ any particle on axis of rotation has $I_i = 0$

I changes as axis of rotation is defined at a different location

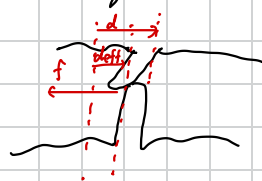
Parallel axis theorem:

$$I_{\text{any pt}} = I_{\text{cm}} + M r_{\perp}^2$$

In some cases,

$I = X M r^2$, where X is some number $0 < X \leq 1$

Modelling Friction:



$\sum F_{\text{net}} = 0$
but $\Delta M \neq 0$
in most cases
 $\text{def} \sim 0.5d$

Chapter 10.

→ Collisions **DOES NOT** require "touching"
 ↳ 2 or more particle interacts with each other
 much more strongly than they interact with the environment.

Momentum is conserved, but not necessarily mechanical energy

3 collision types: **★ helpful in modelling**

★ Elastic collision

- KE conserved
- Ignores sound, heat, plastic deformation

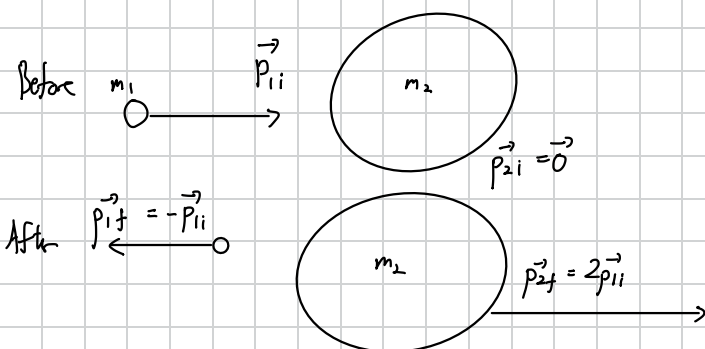
Inelastic collision

- KE changes (decreases)

★ Totally (maximally) inelastic

- Objects stick together
- Loses maximum KE

"Small Ball on Big ball"



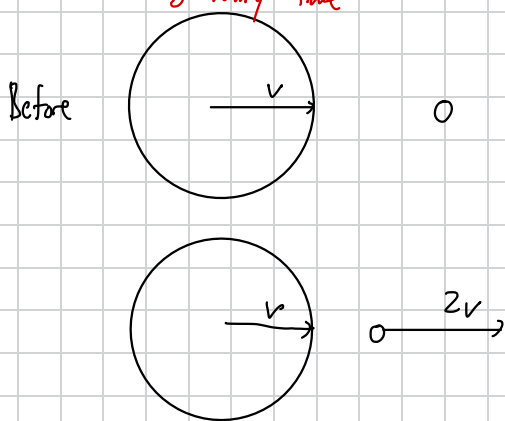
$$\vec{p}_{1i} + \vec{0} = \vec{p}_{1f} + \vec{p}_{2f}$$

$$\vec{p}_{1i} = -\vec{p}_{1f} + \vec{p}_{2f}$$

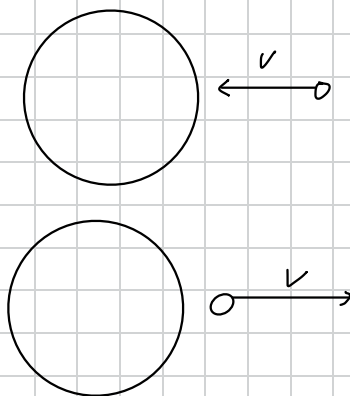
$$2\vec{p}_{1i} = \vec{p}_{2f}$$

Reference frames ("Big ball on small ball")

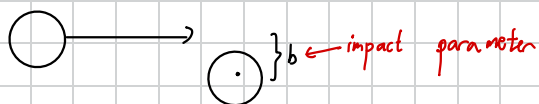
Stationary frame



Reference frame, at v .



3D. collision



Rutherford:

- Random impact parameters
- Conserve momentum
- Found final direction of proton deflection.

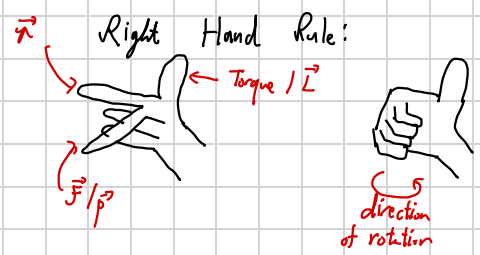
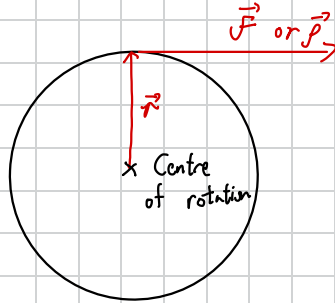
Chapter 11

(Point Particle)

Translational angular momentum:

$$\vec{L}_{\text{trans}} = \vec{r} \times \vec{p}$$

$$|\vec{L}| = |\vec{r}| |\vec{p}| \sin \theta$$



Rotational angular momentum:

$$\vec{L}_{\text{rot}} = I \vec{\omega}, \quad |\vec{L}| = I |\vec{\omega}|, \quad \omega = \frac{v}{R}$$

r : vector from (any) centre to tail of force

K_{rot} and $\frac{1}{2}$

$$K_{\text{rot}} = \frac{1}{2} \omega^2 = \frac{L_{\text{rot}}^2}{2I}$$

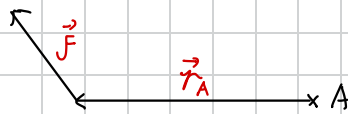
(similarly $K_E = \frac{1}{2} m v^2 = \frac{p^2}{2m}$)

Total angular momentum at point A:

$$\vec{L}_A = \vec{L}_{\text{rot}} + \vec{L}_{\text{trans}, A}$$

Torque:

$$\tau_A = \vec{r}_A \times \vec{F}$$



Conservation of angular momentum:

$\vec{L}$ is conserved when $\tau_{\text{net}} = 0$

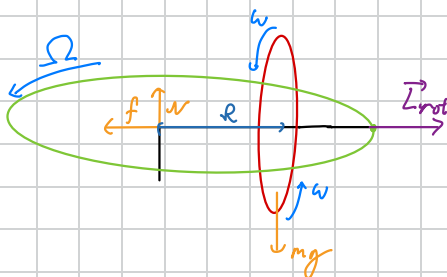
Angular momentum principle:

$$\vec{L}_{f, A} = \vec{L}_{i, A} + \vec{\tau}_{\text{net}, A} \Delta t, \quad \text{update form}$$

$$\frac{d\vec{L}_A}{dt} = \vec{\tau}_{\text{net}, A}, \quad \text{derivative form}$$

$$\Delta L_{A, \text{system}} + \Delta L_{A, \text{surroundings}} = 0, \quad \text{conservation form}$$

Gyroscopes:



We know:

$$N = mg$$

$$\text{Centripetal force} = \frac{mv^2}{R}$$

$$= \frac{m(v\Omega)^2}{R}$$

$$= mR\Omega^2 = f$$

Net translational momentum = 0

Angular momentum principle:

$$\left| \frac{d\vec{L}_{\text{rot}}}{dt} \right| = \Omega L_{\text{rot}} = \tau_{\text{cm}} \rightarrow \Omega L_{\text{rot}} = Rmg$$

$$\tau_{\text{cm}} = Rmg$$

$$\Omega = \frac{Rmg}{L_{\text{rot}}}$$

$$\Omega = \frac{Rmg}{I\omega}$$

Aside: I for point particles:

(Treat particle as mR^2 like // axis theorem)

Aside, calculus and physics

$$v = \frac{dx}{dt}$$

$$\Delta v = \int a \, dt$$

$$a = \frac{dv}{dt} = \frac{dv}{dt}$$

$$\Rightarrow \Delta x = \int \int a \, dt = \int v \, dt$$

$$|\vec{v}| = \frac{|\vec{x}|}{T}$$

$$F_{\text{net}} = \frac{dp}{dt} = \frac{dmv}{dt}, \text{ if } m \text{ is constant} \Rightarrow \Delta p = \int F_{\text{net}} \, dt$$
$$= m \frac{dv}{dt} = m \int a \, dt, \text{ if } a \text{ is constant}$$

$$W = \int \vec{F} \cdot d\vec{r}$$

$$\vec{F} = - \frac{dV}{d\vec{r}}$$

$$P = \frac{dE}{dt} \text{ (and } \vec{F} \cdot \vec{v} \text{)}$$

$$E = \int P \, dt$$

